Data Science PITP Batch 5 Phase 2

Global Fuel Prices Prediction Project Report (2020-2026)

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1. Introduction & Problem Statement

Fuel prices are among the most economically sensitive indicators in the world. They affect transportation costs, inflation, agricultural productivity, and the day to day lives of billions of people. Despite their importance, fuel prices vary dramatically across countries due to a mix of factors: global crude oil benchmarks, regional taxation policies, income levels, and geopolitical dynamics.

This project asks a simple but powerful question: can we predict petrol prices per litre (in USD) for a given country and time period, based on observable economic and structural features? The answer has real world relevance from government policy planning to consumer financial planning.

Using the Global Fuel Prices 2020–2026 dataset, we built and compared three supervised regression models: Linear Regression, Decision Tree Regressor, and Random Forest Regressor. The project covers the complete data science pipeline from raw data ingestion and preprocessing, through feature engineering and exploratory analysis, to model training, evaluation, and interpretation.

2. Dataset Source

The dataset used in this project is the Global Fuel Prices 2020–2026 CSV file, which contains weekly fuel price records for countries around the world. Each row represents one country's fuel pricing snapshot for a given week.

Key columns in the dataset include:

- date the weekly timestamp of the observation
- country & region geographic identifiers
- income_level World Bank income classification (e.g., High, Upper Middle, Lower Middle, Low)
- petrol_usd_liter, diesel_usd_liter, lpg_usd_liter retail fuel prices in USD per litre
- brent_crude_usd the global Brent Crude oil benchmark price
- tax_percentage the share of the retail petrol price attributable to government tax

The dataset is well suited to the problem: it includes both numerical features (prices, percentages) and categorical features (region, income level), spans six years of weekly observations across dozens of countries, and presents a genuine regression prediction challenge.

3. Dataset Processing

Before any analysis could begin, the raw data required careful cleaning and preparation. This stage is often the most unglamorous part of data science, but it is the foundation on which every subsequent step rests.

3.1 Handling Missing Values

A check for null values (df.isnull().sum()) was performed across all columns. Missing values in the three primary price columns petrol_usd_liter, diesel_usd_liter, and brent_crude_usd were imputed using the column mean. This is a standard and reasonable strategy for numerical price data, particularly when the proportion of missing values is small and there is no systematic pattern to the missingness.

3.2 Encoding Categorical Variables

Two categorical columns required conversion to numbers before they could be used in machine learning models. Label Encoding was applied to both region (the geographic region of the country) and income_level (the World Bank income classification). The result was two new numerical columns: region_num and income_num. Label Encoding was appropriate here because these features were used as inputs to tree based models, which do not assume any ordinal relationship between the encoded integers.

3.3 Scaling

After the train test split, StandardScaler was applied to the feature matrix. Scaling transforms all features so they have a mean of zero and a standard deviation of one. This is particularly important for Linear Regression, which is sensitive to feature scale. For tree based models it is less critical, but applying it uniformly ensures a fair comparison.

3.4 Train Test Split

The dataset was divided into a training set (80%) and a testing set (20%) using a fixed random state of 42 for reproducibility. The training set was used exclusively for fitting the models; the test set was held back and used only for final evaluation.

4. Exploratory Data Analysis (EDA)

EDA is the stage where the data begins to tell its story. Rather than jumping straight to modelling, we invested time in understanding the distributions of variables, the relationships between features, and the global patterns in the data.

4.1 Correlation Heatmap

A Pearson correlation heatmap was generated for all numeric columns. The strongest positive correlations were found between brent_crude_usd and petrol_usd_liter confirming the economic intuition that global

crude oil prices are a primary driver of retail petrol prices. The brent_lag_1 feature (explained in Section 5) also showed high correlation with petrol prices, validating its inclusion. tax_percentage showed a moderate positive relationship, reflecting that higher tax countries tend to have higher retail prices.

4.2 Univariate Distributions

Histograms with KDE overlays were produced for petrol_usd_liter, diesel_usd_liter, lpg_usd_liter, brent_crude_usd, and tax_percentage. Petrol and diesel prices showed a right skewed unimodal distribution, with most countries clustering around USD 1.00–1.50 per litre and a long tail toward higher prices (primarily high income, high tax European nations). LPG showed a broader spread. Brent crude displayed the characteristic two peak shape reflecting the COVID 19 crash in 2020 and the price surge through 2021–2022.

4.3 Time Series Trends

A dual axis time series plot tracked global average prices for petrol, diesel, and LPG (left axis) alongside Brent Crude (right axis) from 2020 through 2026. The visualisation clearly captured the collapse in crude prices in early 2020, the recovery through 2021, the sharp spike following the 2022 geopolitical disruptions, and the gradual normalisation thereafter. This time series behaviour reinforced the value of including a lagged crude price feature in the model.

4.4 Country Level Price Comparison

Horizontal bar charts were used to display the 10 cheapest and 10 most expensive countries by median petrol price. Subsidised fuel economies in parts of the Middle East and Central Asia appeared at the cheap end, while densely taxed European nations occupied the expensive end. This visualisation highlighted the enormous heterogeneity in retail fuel prices globally, and reinforced the importance of including income level and region as predictive features.

5. Feature Engineering

Raw features alone are seldom sufficient for strong predictive models. Feature engineering is the craft of transforming and combining raw variables into representations that are more informative for the model.

5.1 Temporal Features

The date column was parsed into a proper datetime object and then decomposed into two numerical features: year and month. This allows the models to capture long term price trends (via year) and seasonal cyclicality (via month), both of which are present in global fuel pricing data.

5.2 Lagged Crude Price (brent_lag_1)

One of the most analytically motivated features in the project was brent_lag_1 the Brent Crude price from the previous week's observation for the same country. Retail fuel prices do not instantaneously adjust to crude price changes; there is a real world lag due to refining, logistics, and pricing contracts. By grouping by country and applying a one period shift, the lagged feature captures this delay. The very first observation per country (which has no prior week) was filled with the global mean crude price.

5.3 Tax Ratio

A tax_ratio feature was derived by dividing tax_percentage by 100 converting the percentage to a decimal proportion. While numerically equivalent to scaling tax_percentage, this makes the feature semantically cleaner (a value of 0.6 means 60% of the retail price is tax) and aids model interpretability.

5.4 Tax Value

The absolute tax component in USD per litre was calculated as tax_val = petrol_usd_liter × (tax_percentage / 100). This feature expresses not just the rate of taxation but its actual monetary weight, which is more directly relevant to price variation across countries.

5.5 Final Feature Set

The final set of features used for model training was: brent_crude_usd, brent_lag_1, tax_percentage, region_num, income_num, year, and month. The target variable was petrol_usd_liter. This combination captures global crude dynamics, fiscal policy signals, geographic and economic context, and temporal patterns.

6. Model Building & Evaluation

Three regression models were trained and evaluated. Each represents a different modelling philosophy and serves as a useful point of comparison.

6.1 Linear Regression

Linear Regression is the simplest and most interpretable baseline model. It assumes a linear relationship between each feature and the target. Given the complexity of global fuel pricing with non linear interactions between income level, region, and crude prices a linear model was not expected to capture all the variance, but provides a meaningful benchmark.

6.2 Decision Tree Regressor

The Decision Tree Regressor partitions the feature space into regions through a series of binary splits, assigning a predicted value to each leaf node. It can capture non linear relationships and interactions between variables but is prone to overfitting when grown to full depth. It was trained with a fixed random_state for reproducibility.

6.3 Random Forest Regressor

Random Forest builds an ensemble of 100 decision trees, each trained on a random bootstrap sample of the training data and a random subset of features. The final prediction is the average of all tree predictions. This averaging process reduces variance dramatically compared to a single decision tree, making Random Forest robust and generally the strongest performer on tabular data problems like this one.

6.4 Evaluation Metrics

All three models were evaluated on the same held out test set using four standard regression metrics: Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R squared (R2 Score).

Model	MAE	MSE	RMSE	R^2 Score
Linear Regression	~0.18	~0.06	~0.24	~0.76
Decision Tree	~0.08	~0.02	~0.14	~0.91
Random Forest ✓Best	~0.05	~0.01	~0.10	~0.96

Table 1 Model Performance Comparison (approximate values; refer to notebook for exact figures)

7. Limitation

The project has several important limitations. First, the model was trained on historical patterns; sudden geopolitical events (such as supply disruptions or sanctions) could produce price movements that fall outside the training distribution. Second, while Label Encoding was used for categorical features, more sophisticated embeddings or one hot encoding might better represent region and income level for linear models. Third, hyperparameter tuning (such as optimising tree depth or number of estimators in Random Forest) was not performed systematically; cross validated grid search could yield further improvement.

8. Conclusion And Insight

Random Forest achieved the lowest error across all three error metrics and the highest R^2 score, indicating that roughly 96% of the variance in petrol prices was explained by the model. The Decision Tree performed substantially better than Linear Regression, confirming that non linear relationships exist in the data. Linear Regression still achieved a respectable R^2 of approximately 0.76, reflecting that the linear component of the relationship (particularly the crude to petrol price link) is genuinely strong.

This project demonstrated the end to end application of data science methods to a real world economic problem. Beginning with raw weekly fuel price data for countries worldwide, we transformed, enriched, and modelled the data to build a system capable of predicting retail petrol prices with high accuracy.

The key findings from this project are:

- Brent Crude oil price is the single strongest predictor of retail petrol prices globally, both in its current form and through the lagged feature.
- Income level and geographic region are important structural predictors, capturing the systematic differences in subsidy regimes and tax policies between countries.
- Feature engineering particularly the brent_lag_1 lag feature meaningfully improved model performance by capturing the real world delay in crude to retail price transmission.
- Random Forest Regressor was the best performing model, outperforming both Linear Regression and Decision Tree across all evaluation metrics.

Dataset Source: Global Fuel Prices 2020–2026 (Kaggle) https://www.kaggle.com/datasets/belbino/global-fuel-prices-20202026

Tools used: Python 3, Pandas, NumPy, Scikit learn, Matplotlib, Seaborn. Environment: Google Colab.

Repository :

https://github.com/AbdurRafay1421/Global-Fuel-Prices-Prediction-Project-Report-2020-2026-